# UWB-based Active SLAM Integrating Uncertainty-Aware Reactive Planning of UAVs

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***Abstract*—Localisation in GNSS-denied indoor environments poses a critical challenge for aerial robotics, for which Ultra-Wideband (UWB) technology offers an effective solution based on accurate ranging measurements. However, range-only Simultaneous Localisation and Mapping (SLAM) techniques inherently suffer poor observability conditions and are subject to geometric ambiguities, specifically the *flip ambiguity* of the trajectory, which can lead to fatal initialisation errors for standard recursive estimation filters like the Extended Kalman Filter (EKF). This work presents a unified Active SLAM framework that addresses these issues by integrating a robust initialisation strategy with reactive trajectory planning. The proposed approach employs a dynamic bank of Weighted Least Squares (WLS) estimators to reliably determine the initial positions of unknown anchors before their inclusion in the filter state, thus following a delayed SLAM approach. In parallel, a hierarchical observability-aware reactive planner modulates the steering command, inducing active manoeuvres required to resolve geometric symmetries and maximise information gain along the path. Experimental validation in a simulated environment demonstrates that the proposed method ensures map consistency and prevents estimation divergence, drastically reducing positioning and mapping errors compared to conventional passive navigation strategies.**

## I. Introduction

Global Navigation Satellite Systems (GNSS) provide reliable localisation in open outdoor environments, but their performance degrades significantly in indoor or cluttered scenarios. Ultra-Wideband (UWB) technology has emerged as an effective alternative for indoor positioning [1], [2] thanks to its robustness to multipath and its ability to provide accurate ranging measurements [3]. When integrated into a probabilistic framework, UWB sensing can be exploited for Simultaneous Localisation and Mapping (SLAM), typically through Bayesian methods or global batch problem, respectively using Extended Kalman Filter SLAM (EKF-SLAM) or graph-based Weighted Least Squares (WLS) [4].

Conventional EKF-SLAM formulations rely on sensors that provide both range and bearing information [4]. This combination constrains the geometry of the problem and enables direct observability of the relative positions between the robot and the landmarks. When only UWB range measurements are available, the problem becomes more challenging. The absence of bearing information leads to a loss of geometric constraints, and the system can become ambiguous without an appropriate initialisation strategy [5]. Initialisation through trilateration is a common approach to recover the initial positions of the unknown entities [6], [7]. However, trilateration based solely on distances suffers from inherent symmetries. In particular, solutions may be mirrored with respect to a reference plane or axis, producing configurations that are mathematically valid but physically inconsistent within the map [8]. If not properly resolved at initialisation, this ambiguity propagates through the SLAM estimation step and compromises the overall consistency of the reconstructed environment [9].

Active trajectory planning driven by the Fisher Information Matrix (FIM) analysis is fundamental for range-only localisation. For example, regarding pure trajectory optimisation, [10] maximises the FIM determinant for target initialisation, demonstrating the superiority of non-causal approaches, while [11] enforces localisability constraints through A- and E-optimality metrics. Moreover, concerning the trade-off between observability and control performance, [12] proposes a multi-objective strategy balancing FIM against energy consumption, whereas [13] addresses the same dilemma by employing a Nonlinear Model Predictive Control (NMPC) framework. Finally, in the context of tightly coupled sensor fusion, [14] applies nonlinear optimisation to jointly estimate the robot state and the unknown anchor position. Beyond trajectory optimisation, the geometric configuration of the infrastructure plays a crucial role in reducing uncertainty [15]. In [16], the authors analyse optimal UWB sensor placement through the Geometric Dilution of Precision (GDoP), enabling dynamic runtime extension of the anchor infrastructure. The challenges of calibration and the resolution of geometric ambiguities, a critical issue in range-only setups, are further addressed in [17], [18], where dedicated procedures are developed to handle symmetries and initialisation in UWB networks.

### A. Problem Formulation and Paper Contribution

We consider an aerial vehicle equipped with an Inertial Measurement Unit (IMU) and UWB ranging sensor. The objective is to localise the Unmanned Aerial Vehicles (UAV), i.e., a drone, and map the environment assuming only partial prior knowledge, limited to a set of four fixed anchors defining the reference frame near the takeoff position. However, range-only SLAM poses a fundamental conflict between navigation efficiency and estimation observability. A trajectory

that minimises travel time, such as a straight line towards the goal, creates a degenerate geometric configuration where the position of the anchors becomes unobservable due to symmetry across the trajectory axis, previously dubbed flip ambiguity. Consequently, reliance on passive estimation filters is insufficient, as the EKF linearisation requires the state estimate to be close to the true value, a condition violated when initialisation is corrupted by such geometric ambiguities. Hence, in this paper, we address the challenges associated with UWB-only EKF-SLAM for UAVs and discuss how appropriate initialisation mechanisms and uncertainty-aware path planning can prevent mirrored solutions and stabilise the estimation process, without relying on external infrastructure or prior knowledge. The combination of EKF-SLAM and uncertainty-aware path planning falls in the realm of Active SLAM solutions.

## II. Proposed Solution

Consider a multi-rotor drone operating in a three-dimensional workspace. The platform is equipped with an IMU and a an UWB tag that is able to collect ranging measurements from a set of UWB anchors deployed in unknown positions in the workspace. The vehicle attitude is assumed to be available from an external orientation estimator, for example through a MARG/Madgwick filter [19]. Given the estimated rotation matrix from the body frame to the world frame, $R(t)$, the linear acceleration expressed in the inertial frame, $a^w$, can be obtained as

$$a^w(t) = R(t)a^b(t) - g, \tag{1}$$

where $a^b$ is the acceleration measured in the body frame by the IMU and $g$ is the gravity vector. The UAV is also capable of measuring its distance from the set of UWB anchors. The anchors are stationary, but the reception of the ranging signal is physically constrained by the maximum operating range of the UWB modules, denoted as $r_{\max}$. Consequently, a measurement about the $i$-th anchor at time $t$ is modelled as

$$z_i(t) = \|X(t) - p_i\| + \eta_i, \text{ if } \|X(t) - p_i\| \leq r_{\max}, \tag{2}$$

where $p_i$ is the position of the $i$-th anchor, while the measurement uncertainty is assumed to be white and normally distributed, i.e., $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$.

In this setup, we distinguish between two types of anchors: those with known positions, denoted by the set $\mathcal{A}^{\text{fix}}$, and those whose positions are unknown (estimated by SLAM), denoted by the set $\mathcal{A}^{\text{slam}}$. This differentiation implies two distinct measurement models: $z_i^{\text{fix}}$ for anchors $p_i^{\text{fix}} \in \mathcal{A}^{\text{fix}}$, where $p_i^{\text{fix}}$ is treated as a known constant and are used to firstly localise the robot in the starting position: those anchors are four and placed as the vertex of a quadrilateral, which are the most effective locations to reduce the localisation uncertainty [20]. Instead, $z_i^{\text{slam}}$ are the measurements for the anchors $p_i^{\text{slam}} \in \mathcal{A}^{\text{slam}}$, where $p_i^{\text{slam}}$ is treated as a variable state component of the EKF. Specifically, the SLAM anchors are assumed to be ground-based stations positioned at a predefined height; therefore, their vertical component $z$ is considered constant, limiting the estimation to the planar $(x, y)$ coordinates. Figure 1 shows the deployment of the anchors in the operational space.

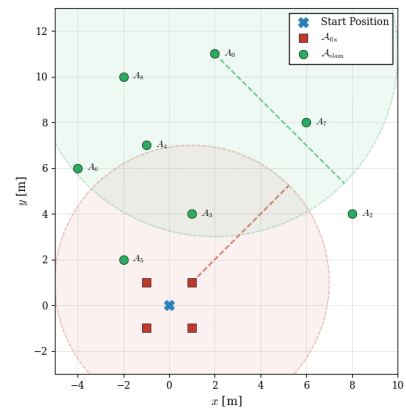



Fig. 1: Simulated Environment for UWB Localisation and SLAM. The figure presents the simulated scenario from an overhead perspective. The fixed reference anchors with known positions, $\mathcal{A}^{\text{fix}}$, are represented by red squares. The SLAM anchors with unknown $(x, y)$ coordinates, $\mathcal{A}^{\text{slam}}$, are shown as green circles. The blue cross marks the drone's initial takeoff point. The shaded areas represent the sensing ranges, uniformly set to 8 m for all SLAM anchors and 6 m for all fixed anchors.

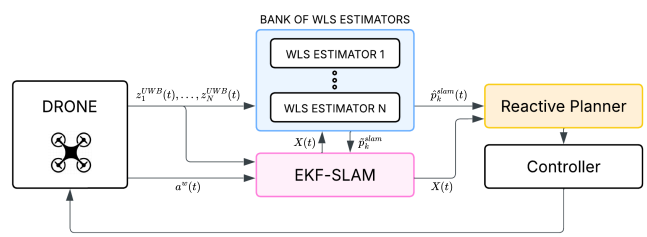



Fig. 2: Architecture of the proposed framework.

### A. Framework Overview

The proposed autonomous drone framework is structured around three fundamental modules: the bank of WLS Estimators, the EKF-SLAM filter, and the Reactive Planner (RP), as illustrated in Figure 2. While navigating, the drone continuously collects proper acceleration measurements $a^w(t)$ and UWB ranging measurements. The system distinguishes between anchors with known positions, denoted as $\mathcal{A}^{\text{fix}}$, and unknown environmental anchors in $\mathcal{A}^{\text{slam}}$ that need to be mapped. Measurements associated with the latter set are processed by a dynamic bank of WLS estimators, which provides time-varying position estimates $\hat{p}_i^{\text{slam}}(t)$. Once predefined reliability thresholds are satisfied, the current WLS estimate is used to initialise the corresponding feature within the EKF state. Crucially, the RP exploits the pre-convergence estimates $\hat{p}_i^{\text{slam}}(t)$, together with the current system state $X(t)$, to steer the drone along trajectories that reduce the uncertainty on the

anchor initialization, thereby guaranteeing robust estimation and implementing an Active SLAM strategy.

### B. EKF-SLAM

The core module of our framework employs an Extended Kalman Filter (EKF) to jointly estimate the robotic platform motion and the positions of the environmental anchors. The full state vector $x$ is formulated within an EKF-SLAM structure, composed of the platform 3D kinematic state position $X$, velocity $v$, and acceleration $a$ vectors, and the vector of the $M$ anchor positions, where the $i$-th anchor position is estimated in 2D as $p_i^{\text{slam}} = [p_{i.x}^{\text{slam}}, p_{i,y}^{\text{slam}}]^T$. The full state vector is then $x = [X^T, v^T, a^T, {p_1^{\text{slam}}}^T, \cdots, {p_M^{\text{slam}}}^T]^T$. The total dimension of the state vector is $9 + 2M$. The anchors are initialised inside of the state once the WLS Estimator computes the initial estimate (see Section II-C).

*1) Prediction:* The EKF is executed every $\Delta t$ seconds in a discrete-time fashion, hence the state transition matrix $\Phi$ propagates the state forward at every time step $\Delta t$. $\Phi$ and the corresponding process noise covariance matrix $Q$, shares the same block-diagonal form, i.e.

$$\Phi = \begin{bmatrix} \Phi_{\text{drone}} & 0 \\ 0 & \Phi_{\text{a}} \end{bmatrix}, \; Q = \begin{bmatrix} Q_{\text{drone}} & 0 \\ 0 & Q_{\text{a}} \end{bmatrix}.$$

The block diagonal components of the transition matrix $\Phi$, i.e., $\Phi_{\text{drone}}$ for the constant acceleration model of the drone and the $2M \times 2M$ anchor transition block $\Phi_{\text{a}}$ for the static anchors, are explicitly defined as:

$$\Phi_{\text{drone}} = \begin{bmatrix} I_3 & I_3\Delta t & I_3\frac{\Delta t^2}{2} \\ 0_3 & I_3 & I_3\Delta t \\ 0_3 & 0_3 & I_3 \end{bmatrix}, \; \Phi_{\text{a}} = I_{2M},$$

where $I_n$ corresponds to the $n \times n$ identity matrix and $0_n$ represents the $n \times n$ zero matrix. The corresponding uncertainty covariance block $Q_{\text{drone}}$ (jerk uncertainty model) and the anchor uncertainty block $Q_{\text{a}}$ (conditional random walk model) are given by:

$$Q_{\text{drone}} = \sigma_d^2 I_3 \otimes \begin{bmatrix} \frac{\Delta t^5}{20} & \frac{\Delta t^4}{8} & \frac{\Delta t^3}{6} \\ \frac{\Delta t^4}{8} & \frac{\Delta t^3}{3} & \frac{\Delta t^2}{2} \\ \frac{\Delta t^3}{6} & \frac{\Delta t^2}{2} & \Delta t \end{bmatrix}, Q_{\text{a}} = I_M \otimes (\sigma_p^2 \Delta t I_2),$$

where $\sigma_d^2$ is the variance of the uncertainty induces by the estimates of the accelerations. Moreover, $\sigma_p^2$ represents, albeit defined generally here, the uncertainty conditionally applied to the anchors to simulate a random walk, hence to enhance the estimates: in general, it is supposed to be less than $\sigma_\eta^2$ affecting the ranging measurements and defined in (2). Finally, '$\otimes$' is the Kronecker product.

*2) Update:* The EKF performs two separate updates based on the measured information from the IMU and the UWB anchors. The IMU data, provides the acceleration in the global reference frame $a^w(t)$ in (1). This information is used as a pseudo-measurement to update the acceleration vector already present in the state. The $H^{\text{IMU}}$ is simply the matrix that linearly relates the measured acceleration to the full state vector $x$, given by:

$$H^{\text{IMU}} = \begin{bmatrix} 0_3 & 0_3 & I_3 & 0_{3\times 2M} \end{bmatrix},$$

whose uncertainty is related to $\Sigma_a$, i.e., the covariance matrix of the IMU measurements on the three IMU axes.

Regarding the ranging, the UWB sensor update is based on two distinct classes of measurements: those originating from anchors with known (fixed) positions in $\mathcal{A}^{\text{fix}}$ and those originating from anchors whose positions are actively estimated within the state vector $x$, i.e., those in $\mathcal{A}^{\text{slam}}$. The full UWB measurement vector, $Z^{\text{UWB}}(t)$, is formed by concatenating the range measurements from the fixed and SLAM anchor sets. Let $N_{\text{fix}}$ and $N_{\text{slam}}$ be the number of measurements received from these sets, respectively. The full vector is $Z^{\text{UWB}}(t) = [z_1^{\text{fix}}, \cdots, z_{N_{\text{fix}}}^{\text{fix}}, z_1^{\text{slam}}, \cdots, z_{N_{\text{slam}}}^{\text{slam}}]^T$, where the measurements $z_i^{\text{fix}}$ are used for the drone localisation only (when available), while the $z_i^{\text{slam}}$ are used as a SLAM input. The predicted measurement $h(x)$ for a generic anchor $i$ is the Euclidean distance between the robot position $X$ and the $i$-th anchor position $p_i$ (either in $\mathcal{A}^{\text{fix}}$ or in $\mathcal{A}^{\text{slam}}$) is $h_i(x) = \|X - p_i\|$, whose linearised measurement model $H$ is therefore also composed of two different parts:

$$H^{\text{UWB}} = \begin{bmatrix} H^{\text{fix}} \\ H^{\text{slam}} \end{bmatrix},$$

where $H^{\text{fix}}$ and $H^{\text{slam}}$ are obtained by stacking the $N_{\text{fix}}$ and the $N_{\text{slam}}$ rows, respectively, of the Jacobian matrix rows, given by:

$$H_i^{\text{fix}} = \begin{bmatrix} u_i^T & 0_{1\times 3} & 0_{1\times 3} & 0_{1\times 2M} \end{bmatrix},$$
$$H_j^{\text{slam}} = \begin{bmatrix} u_j^T & 0_{1\times 3} & 0_{1\times 3} & 0 \cdots - s_j^T \ldots 0 \end{bmatrix},$$

where $u_k = (p - p_k)/\|p - p_k\|$ is the unit vector that corresponds to the direction between the robot and the anchor taking the measurement, while $s_j^T$ is the derivative with respect to the 2D estimated anchor position. $H_i^{\text{fix}}$ and $H_j^{\text{slam}}$ are evaluated at the robot state $x$ after the prediction step.

### C. WLS Estimator

To improve the performance of the EKF, a reliable initial estimate of the anchor position in $\mathcal{A}^{\text{slam}}$ is needed. To this end, we employ a dynamic bank of WLS estimators. A new estimator instance is initialised sequentially as soon as the system detects the first range measurement from a previously unobserved anchor ID. Each estimator operates on a sliding window of range measurements. While the measurement model follows (2), the weighting scheme must account for the uncertainty in the robot trajectory. The variance $\Sigma_i$ for the $i$-th anchor is therefore computed as:

$$\Sigma_i(k) = \sigma_\eta^2 + H_i(k) P_{\text{Robot}}(k) H_i(k)^T, \tag{3}$$

where $P_{\text{Robot}}(k)$ is the robot estimation error covariance matrix at time-step $k$ as estimated by the EKF, while $H_i(k) =$

$u_i(k)^T$ is the Jacobian direction vector. To bound the computing cost and ensure numerical stability, we do not utilise all available raw data. Instead, we leverage on the relationship between the FIM and the GDoP, hence the FIM assesses the reliability of the solution of the WLS. Since for range-only measurements, the Jacobian matrix $H$ is composed of the unit line-of-sight vectors $u_i$ pointing from the anchors to the estimate, using the weight matrix $\Sigma$, whose elements are given in (3), the FIM $J$ for each anchor is derived as the weighted sum of the outer products of these direction vectors:

$$J = H^T \Sigma^{-1} H = \sum_{k=1}^{M} w_k u_k u_k^T,$$

where $w_k = \Sigma_i(k)^{-1}$ is the weight of the $k$-th measurement and $M$ is the total number of measurements. According to the Cramér-Rao Lower Bound (CRLB), the covariance of the estimation error is bounded by $J^{-1}$. This relationship explicitly links uncertainty to geometry: if the vectors $u_i$ are collinear or span a narrow sector, $J$ becomes ill-conditioned, resulting in high uncertainty, a phenomenon known as GDoP. Beyond local uncertainty, range measurements suffer from global *flip ambiguity*: if the robot trajectory is linear (in 2D) or planar (in 3D), the true position becomes mathematically indistinguishable from its symmetric counterpart mirrored across the path, leading to a singular configuration [17], [18]. Hence, the system filters the incoming stream of ranging measurements to retain the subset whose direction vectors $u_k$ maximise the angular span within the sliding window. This selection strategy prevents the processing of redundant, collinear data that would lead to an ill-conditioned FIM, ensuring a robust initialisation even with a limited buffer size. To summarise, the $i$-th anchor position estimate $\hat{p}_i^{\text{slam}}$ is accepted and passed to the EKF only when three validation criteria, namely stable variance of the estimates, no flipping ambiguity, and geometric FIM criteria satisfied, simultaneously satisfy pre-defined thresholds.

### D. Information-Aware Planner Design

To achieve Active SLAM, the planner implements a potential field method and relies on two specialised force vectors, driving the robot along certain directions, derived from the observability analysis: one to resolve the geometric ambiguities and another to maximise the information gain.

*1) Symmetry-Breaking Force:* To quantify the ambiguity of an estimated anchor $i$, we define the residual of the $k$-th measurement related to that anchor as $r_k(\hat{p}_i^{\text{slam}}) = z_i(k) - \|X(t_k) - \hat{p}_i^{\text{slam}}\|$ and the WLS cost function

$$\mathcal{J}(\hat{p}_i^{\text{slam}}) = \sum_{k=1}^{M} r_k(\hat{p}_i^{\text{slam}})^T \Sigma_i^{-1} r_k(\hat{p}_i^{\text{slam}}).$$

Similarly we define the same quantities with respect to the mirrored (flipped) quantities $\mathcal{J}(\hat{p}_i^{\text{slam,mir}})$. Then, we utilise the metric $\mathcal{J}_i^{\text{ratio}} = \frac{\mathcal{J}(\hat{p}_i^{\text{slam}})}{\mathcal{J}(\hat{p}_i^{\text{slam,mir}})}$: a value close to unity indicates a critical flipped configuration. We then use the following novel result, whose proof is omitted for brevity.

**Proposition 1.** *Given a linear trajectory defined by the principal direction $v$, the local exploration direction $u^*$ that maximises the expected gradient of the distinguishability ratio $\mathcal{J}_i^{\text{ratio}}$ is orthogonal to $v$.*

Hence, we can define the force direction, denoted as $v_\perp$, as the unit vector orthogonal to $v$. Applying a force in this direction induces the lateral manoeuvre required to avoid the flipped conditions.

*2) Information-Maximising Force:* When ambiguity is low, the objective shifts to reducing the covariance of the estimate. According to the E-Optimality criterion [21], the uncertainty ellipsoid in range-only localisation is best reduced by measurements that maximise angular diversity. Geometrically, this corresponds to moving tangentially to the circle centred at the estimated anchor position $\hat{p}_i^{\text{slam}}$. We therefore define the Information Force, denoted as $F_i$, as the tangent vector orthogonal to the line-of-sight segment connecting the robot and the anchor.

*3) Reactive Planning Strategy:* The proposed framework integrates the *Symmetry-Breaking Force* and the *Information-Maximising Force* into a hierarchical reactive planner that arbitrates between the global goal and the estimation integrity requirements. The planner computes the steering command $u_{cmd}$ based on the following priority levels:

*Priority 1: Ambiguity Resolution:* The highest priority is assigned to preserving map consistency. If any active anchor $i$ exhibits high ambiguity (i.e., $\mathcal{J}_i^{\text{ratio}} < \Gamma_{thresh}$), the planner overrides the nominal path. The steering command is computed by blending the goal direction $g$ with the symmetry-breaking force:

$$u_{raw} = g + w(\mathcal{J}_i^{\text{ratio}}) \cdot v_\perp, \tag{4}$$

where the weight $w$ is inversely proportional to $\mathcal{J}_i^{\text{ratio}}$, ensuring that severe ambiguities result in stronger lateral deviations.

*Priority 2: Information-Aware Blending:* If the map is consistent, the planner seeks to improve its accuracy. It evaluates the alignment between the goal $g$ and the information force $F_i$. We employ an angular blending function $\alpha(\theta)$, where $\theta$ is the angle between $g$ and $F_i$:

$$u_{raw} = (1 - \alpha(\theta))g + \alpha(\theta)F_i. \tag{5}$$

Therefore, the system prioritises the tangent direction ($\alpha \approx 1$) only when it is roughly aligned with the goal direction ($\theta < \theta_{min}$), allowing for "efficient exploration." Conversely, if the directions are contrasting, the goal takes precedence to prevent inefficient detours.

*Kinematic Smoothing:* Finally, to generate feasible trajectories for the drone, the raw command $u_{raw}$ undergoes a low-pass filter and a turn-rate limiter, smoothing abrupt transitions between the different behavioural modes.

## III. Experimental Results

To validate the proposed Active SLAM framework, we conducted extensive simulations using the architecture described in Section II. The experiments aim to evaluate the

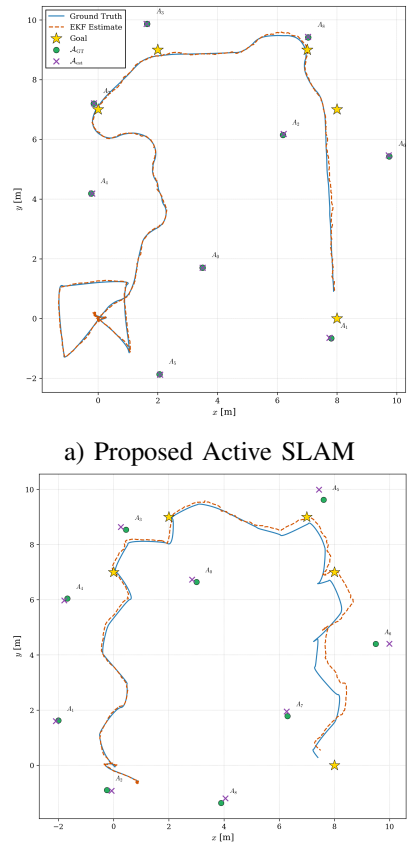
TABLE I: Comparison of mapping and localisation accuracy over Monte Carlo simulations (mean $\pm$ standard deviation).

| Metric | Baseline planner | Active SLAM (Ours) |
|---|---|---|
| Avg. Anchor Error [m] | $0.16 \pm 0.15$ | $\mathbf{0.10 \pm 0.17}$ |
| Robot Pos. RMSE [m] | $0.3 \pm 0.23$ | $\mathbf{0.21 \pm 0.29}$ |
| Robot Pos. RMSE $x-y$ [m] | $0.19 \pm 0.14$ | $\mathbf{0.13 \pm 0.20}$ |
| Robot Pos. RMSE $z$ [m] | $0.22 \pm 0.19$ | $\mathbf{0.16 \pm 0.21}$ |
| Failure rate [%] | 4.0 | **2.0** |

proposed Active SLAM framework through extensive simulations in a PX4 SITL environment integrated with ROS 2. The full project can be downloaded at www.github.com/davidenascivera/UWB-Obs-Aware-SLAM. A Monte Carlo evaluation was performed over $N = 100$ independent runs, each with randomised anchor positions and randomised initial UAV positions. The UWB measurements were modelled with a standard deviation of 0.15 m, and the threshold for the $\mathcal{J}^{\text{ratio}}$ was set to $\Gamma_{thresh} = 1.5$ as a representative value to discriminate near-symmetric configurations from well-conditioned geometries. The inertial data were generated directly by the PX4 SITL dynamics. All experiments were executed on a Dell XPS 15 equipped with an Intel Core i7 8th generation processor and 16 GB of memory.

### *A. Monte Carlo Evaluation of Localisation and Mapping Performance*

To assess the robustness of the proposed approach, the results of the Monte Carlo simulations are analysed by comparing the proposed uncertainty-aware reactive planner (exemplified in Figure 3(a)) against a baseline navigation strategy based on a baseline zig-zag motion pattern (Figure 3(b)), which provides sufficient lateral excitation to avoid degenerate motion patterns (e.g., straight line motions) but does not actively minimise the estimation uncertainty. This improvement is quantitatively reported in Table I: the proposed method reduces the overall robot position RMSE from 0.3 m to 0.21 m on average, and similarly improves both the horizontal $x-y$ and vertical $x$ localisation accuracy. In addition, the failure rate is halved, highlighting the increased robustness of the proposed active strategy. As the number of anchors increases, the performance difference between the methods naturally decreases, due to the improved geometric observability of the system, but the proposed method consistently has higher performance with respect to the baseline. Moreover, for both methods the localisation uncertainty is typically more pronounced along the vertical $z$ axis. This behaviour is expected in our experimental setup, since the anchors are deployed at the same height, yielding a near-coplanar geometry that provides weaker observability in the direction orthogonal to the anchor plane (see Table I). For what concerns the mapping of the anchors, Figure 4 reports the time evolution of the estimation error of each anchor provided by the proposed method. To summarise, the proposed strategy achieves consistently lower localisation and mapping errors, as well as a reduced failure rate, over the Monte Carlo simulations.




a) Proposed Active SLAM

b) Baseline Planner

Fig. 3: Comparison of trajectory behaviours. Top (a): The Active SLAM (blue) maintains a consistent state estimate along the trajectory. Bottom (b): The baseline planner exhibits a progressive degradation of the state estimation as the robot moves farther from the initial position, resulting in increasing localisation and mapping errors.

### *B. Computational Complexity*

The computational complexity of the proposed framework is dominated by the EKF-SLAM backend, while the additional components introduced by the active strategy generate limited overhead. Each WLS-based anchor initialiser operates on a fixed-size sliding window of measurements. Both the linear initialisation and the subsequent Gauss–Newton refinement scale linearly with the window size, yielding a per-estimator complexity of $O(n)$. Considering $m$ concurrently active estimators, the overall WLS bank complexity is $O(mn)$. The information-aware APF planner evaluates candidate motions by iterating over the set of active estimators. Since each evaluation involves only constant-time operations, the planner complexity scales linearly as $O(m)$. The EKF-SLAM prediction and update steps exhibit the standard cubic complexity $O(d^3)$ with respect to the state dimension $d$. In the considered

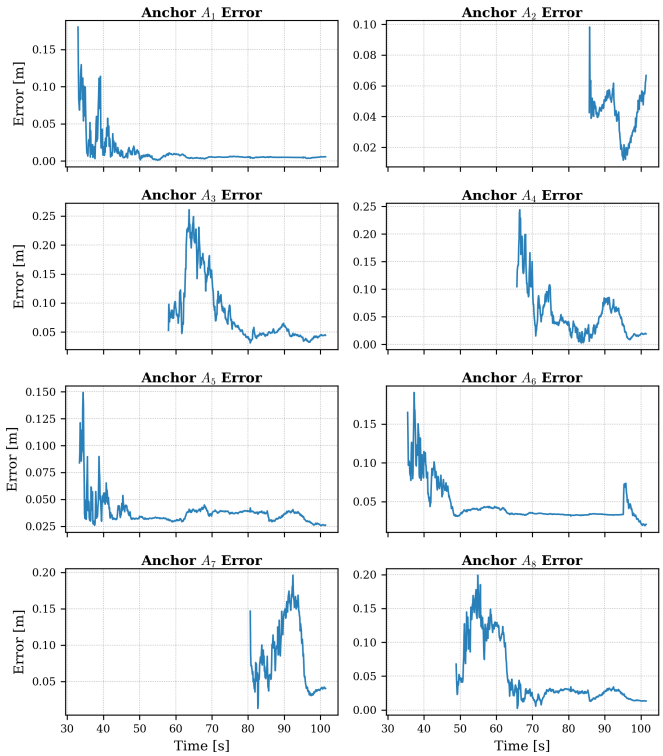



Fig. 4: Temporal evolution of the estimation error for the unknown anchors of Figure 3(a) using the Active SLAM framework. The error is undefined prior to the WLS initialisation event.

setup, $d = 9 + 2n_{slam}$ is bounded by the number of initialised anchors. Despite being implemented in Python, the proposed framework achieves real-time execution within the PX4 SITL simulation environment on the considered desktop platform.

## IV. Conclusion

This work presented an Active SLAM framework for range-only navigation using a hierarchical reactive controller. The core contribution is a strategy that integrates observability metrics into the control loop to resolve geometric ambiguities. Experimental results demonstrate that standard linear trajectories lead to mirrored anchor initialisations, which subsequently cause the navigation filter to diverge due to conflicting measurement updates. Conversely, the proposed active approach generates information-rich trajectories that ensure map consistency and precise localisation. Future research will focus on relaxing the requirement for fixed reference anchors, investigating self-initialisation strategies to bootstrap the SLAM process independently. A systematic sensitivity analysis is left for future work.

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